| Part | Content |
|---|---|
| Problem | The graphs of $y = x^3 - 3x + 2$ and $x + 4y = 4$ intersect in the points $(x_1, y_1)$, $(x_2, y_2)$, and $(x_3, y_3)$. If $x_1 + x_2 + x_3 = A$ and $y_1 + y_2 + y_3 = B$, compute the ordered pair $(A, B)$. |
| Noisy Prefix | From $x + 4y = 4$, the prefix derives $$y = 1 - \frac{x}{4}.$$ Substituting this into $y = x^3 - 3x + 2$ gives $$1 - \frac{x}{4} = x^3 - 3x + 2,$$ and hence $4x^3 - 11x + 4 = 0$. The prefix then attempts to find the individual roots using the Rational Root Theorem, testing candidates such as $x = 1$, $x = \frac{1}{2}$, $x = -\frac{1}{2}$, $x = 2$, and $x = -1$. [omitted: several direct substitutions showing that these candidates are not roots.] The prefix keeps searching for explicit roots, although the problem only asks for the sums of the coordinates. |
| Model Continuation | **It seems the Rational Root Theorem isn't yielding roots easily. Let's try another method.** Given $4x^3 - 11x + 4 = 0$, the continuation writes it as $$4x^3 + 0x^2 - 11x + 4 = 0.$$ **By Vieta's formulas**, $$x_1 + x_2 + x_3 = -\frac{0}{4} = 0,$$ so $A = 0$. Using $y = 1 - x/4$, $$y_1 + y_2 + y_3 = \left(1 - \frac{x_1}{4}\right) + \left(1 - \frac{x_2}{4}\right) + \left(1 - \frac{x_3}{4}\right) = 3 - \frac{x_1 + x_2 + x_3}{4} = 3.$$ Thus, the final answer is $\boxed{(0, 3)}$. |